Macroeconomic Shocks and Equity Market Dynamics in Vietnam: A VAR Analysis

Nguyen Thi Van Anh

DSEB-MFE, National Economics University
Course: Time Series Analysis
Student ID: 11230603

Abstract–**This paper examines the macroeconomic determinants of VN-Index returns from January 2010 to December 2024 using a six-variable VAR(2) model with structural shocks identified through Cholesky decomposition. The empirical results indicate that among the selected macroeconomic variables, the U.S. Federal Funds Rate exerts the most significant influence on VN-Index movements, while the stock market remains largely driven by its own internal dynamics. Variance decomposition and impulse response analysis further suggest that external monetary conditions represent the primary identifiable macroeconomic transmission channel to the Vietnamese stock market.**

Keywords: **VN-Index, Vector Autoregression, Impulse Response Function, Variance Decomposition, Federal Funds Rate, Emerging Markets.**

I. INTRODUCTION

Stock markets serve as a critical conduit for capital formation and resource allocation, linking savers with productive investment opportunities while also reflecting broader macroeconomic conditions. In emerging economies, where financial systems are still developing, equity markets play an especially important role in mobilising long-term capital for both firms and governments. In Vietnam, the VN-Index – the benchmark index of the Ho Chi Minh Stock Exchange (HOSE) – has expanded substantially since its establishment in 2000, accompanied by rapid growth in listed firms, market capitalisation, and investor participation. By 2024, the Vietnamese stock market had become an increasingly important component of the national economy, with market capitalisation representing a significant share of GDP. Beyond its financing role, the VN-Index also functions as a real-time indicator of macroeconomic sentiment. Episodes such as the COVID-19 shock and the U.S. Federal Reserve tightening cycle demonstrated the market's sensitivity to both domestic and external macroeconomic disturbances. Understanding the macroeconomic drivers of VN-Index fluctuations is therefore important for investors managing systematic risk, policymakers monitoring financial stability, and researchers studying shock transmission in an emerging open economy.

The theoretical foundation for a macroeconomic approach to equity pricing rests on the Arbitrage Pricing Theory (APT) of [1], as operationalised by [2], who demonstrate that systematic macroeconomic variables – rather than a single market factor – are the fundamental drivers of equity returns. Under this framework, if asset prices rationally reflect expectations about future economic activity, then any variable that generates systematic, undiversifiable risk must be priced into equity returns. [2] identify five such channels. First, changes in interest rates alter the discount rate applied to future corporate cash flows: a rise in rates reduces the present value of expected earnings, depressing equity prices [3]. Second, inflation erodes the real value of nominal returns and amplifies uncertainty over future cash flows; critically, it is the unexpected component of inflation that carries significant pricing power, since expected inflation is already embedded in nominal discount rates [4]. Third, exchange rate movements affect the international competitiveness of domestically listed firms – currency depreciation improves export margins in the short run but may signal broader macroeconomic instability [5]. Fourth, oil price shocks transmit supply-side cost pressures directly to corporate profitability: higher energy costs compress margins, reduce expected dividends, and lower equity valuations [6]. For a small, open, emerging economy such as Vietnam, however, a fifth channel – external monetary conditions – may dominate all domestic factors. [7] demonstrate that international economic forces are priced into equity markets even under conditions of market segmentation, operating through real-economy linkages such as trade flows and cross-border capital movements. More directly, [3] show that unexpected changes in the U.S. Federal Funds Rate shift the equity risk premium and alter investor risk appetite globally, while [8] confirm that Fed tightening generates significant capital outflow pressures in emerging markets – effects that were particularly pronounced during the 2022–2023 hiking cycle. Taken together, these channels motivate the inclusion of FFR, oil price, CPI, deposit rate, and NEER as candidate macroeconomic determinants of VN-Index returns in the present study.

Despite growing interest in the macroeconomic determinants of emerging market equity returns, empirical evidence specific to Vietnam remains limited. Existing studies either focus on monetary policy transmission to output and inflation rather than to equity prices [9], or examine only a subset of the analytical toolkit available within the VAR framework [5]. Crucially, no existing study combines Granger causality analysis, orthogonalized impulse response functions, forecast error variance decomposition, and out-of-sample forecast evaluation within a unified VAR framework applied specifically to VN-Index returns. Furthermore, the sample periods of prior work predate two structurally important episodes: the COVID-19 shock of 2020 and the Federal Reserve's aggressive tightening cycle of 2022–2023, during which the FFR rose from near zero to 5.5% and triggered significant capital outflows from emerging markets [8]. Whether these episodes altered the relative importance of domestic versus external macro channels for Vietnamese equities remains an open empirical question. Against this backdrop, this paper asks: *Which macroeconomic variables drive VN-Index returns over 2010–2024, through what channels do they operate, and how accurately can a VAR model forecast Vietnamese equity market movements?*

This paper contributes to the literature on macroeconomic determinants of emerging market equity returns in three ways. First, it provides new evidence on the relative importance of domestic and external macroeconomic forces in Vietnam over a period encompassing both the COVID-19 shock and the 2022–2023 Federal Reserve tightening cycle, showing that U.S. monetary policy, proxied by the Federal Funds Rate, dominates domestic monetary variables in explaining VN-Index dynamics. Second, it employs a unified VAR framework that jointly integrates Granger causality tests, impulse response functions, forecast error variance decomposition, and out-of-sample forecasting, allowing for a consistent evaluation of both structural transmission mechanisms and predictive performance, which has been rarely combined in the existing Vietnam-focused literature. Third, the results document a high degree of idiosyncratic variation in VN-Index returns, with own shocks accounting for the majority of forecast error variance, highlighting the limited role of macroeconomic fundamentals and providing new evidence on the stage of informational efficiency and market development in Vietnam.

The remainder of this paper is organised as follows. Section 2 reviews the theoretical background and empirical literature on the macroeconomic determinants of stock market returns. Section 3 introduces the data, variable construction, and empirical methodology employed in the analysis. Section 4 presents the main empirical findings, including the VAR estimation results, impulse response analysis, variance decomposition, and forecasting performance. Section 5 discusses the broader economic implications of the findings for investors and policymakers. Finally, Section 6 concludes the paper.

II. LITERATURE REVIEW

A. *Theoretical Background: APT and Macroeconomic Risk Factors*

The theoretical foundation for a macroeconomic approach to equity pricing rests on the Arbitrage Pricing Theory (APT) proposed by [1], which posits that expected equity returns are determined by multiple systematic risk factors rather than the single market factor of the Capital Asset Pricing Model [10]. Under APT, any variable that generates pervasive, undiversifiable risk must be compensated through expected returns; the key empirical challenge is to identify which macroeconomic variables constitute such systematic factors.

[2] provide the most influential empirical operationalisation of APT, testing whether innovations in macroeconomic variables are priced risks in the U.S. stock market over 1953–1983. Their paper tests whether innovations in macroeconomic variables are risks that are rewarded in the stock market, identifying industrial production, unexpected inflation, the term structure of interest rates, and the spread between high- and low-grade bonds as significantly priced sources of risk. These five variables represent the primary channels through which the real economy transmits information to equity valuations: changes in interest rates alter the discount rate applied to future cash flows; inflation erodes the real value of nominal returns; and industrial production proxies for aggregate economic activity and expected corporate earnings. The [2] framework motivates the inclusion of CPI, deposit rate, NEER, oil price, and FFR as candidate macroeconomic determinants of VN-Index returns in the present study, with each variable representing a distinct channel of systematic risk transmission.

The efficient market hypothesis [11] provides a complementary theoretical perspective: under EMH, stock prices fully and instantaneously reflect all publicly available information, including macroeconomic announcements. A direct implication is that only *unexpected* components of macroeconomic changes should generate equity price responses, since expected changes are already embedded in current valuations. [4] demonstrates that the negative stock-return–inflation relationship documented in the empirical literature is a spurious reflection of the positive correlation between real activity and equity returns: anticipated inflation is priced in, while unexpected inflation generates real return shocks. This distinction between anticipated and unanticipated macroeconomic changes underlies the impulse response methodology adopted in the present paper.

B. *Monetary Policy and Equity Markets: Evidence from Developed Markets*

The relationship between monetary policy and equity market returns has attracted substantial empirical attention, particularly following the work of [3], who provide a rigorous decomposition of how Federal Reserve policy affects U.S. equity prices. A central methodological contribution of their paper is the separation of FFR changes into anticipated and unanticipated components using Federal funds futures contracts, on the grounds that financial markets respond only to the surprise component of policy actions. Their event-study results show that an unexpected 25-basis-point reduction in the FFR is associated with approximately a 1% increase in broad U.S. equity indices, while anticipated rate changes generate no significant market response.

More importantly for the present study, [3] employ a VAR decomposition – analogous in structure to the methodology adopted here – to identify the economic channels through which FFR shocks affect equity prices. Their principal finding is that monetary policy influences equity valuations primarily through the *equity risk premium channel* rather than through changes in real interest rates: an unexpected FFR tightening raises the required return on risky assets, compressing equity valuations even when the effect on real discount rates is transitory. This finding has important implications for emerging markets such as Vietnam, where the equity risk premium channel may operate through international capital flows rather than domestic discount rate adjustments.

Complementary evidence from the international monetary spillover literature supports this mechanism. [8] examine the spillover effects of U.S. monetary policy on emerging markets during periods of uncertainty, finding that FFR tightening generates significant capital outflow pressures in emerging economies through the portfolio rebalancing and risk appetite channels. Their results suggest that the magnitude of these spillovers is amplified during rapid hiking cycles – precisely the episode represented by the 2022–2023 period in the present study's sample.

C. *Macroeconomic Determinants of Stock Returns in Emerging Markets*

The transmission of macroeconomic shocks to equity prices in emerging markets differs systematically from the patterns observed in developed economies, reflecting structural differences in market size, investor composition, liquidity, and institutional quality. [7] demonstrate that international economic forces are priced into equity markets even under conditions of market segmentation, with the degree of integration depending on the openness of the economy to international capital flows. For small, open emerging economies such as Vietnam, this implies that external monetary conditions may exert a stronger influence on equity valuations than domestic macro fundamentals.

The oil price channel represents a further dimension of external shock transmission that is particularly relevant for emerging market economies. [6] documents significant negative responses of equity returns to oil price increases in the U.S. market, operating through the cost-side compression of corporate profit margins and the adverse effect of energy price uncertainty on investment decisions. For oil-importing emerging economies, these cost-channel effects are compounded by the balance-of-payments pressures generated by higher import bills, which can trigger exchange rate depreciation and inflationary spillovers.

D. *Prior Evidence on the Vietnamese Stock Market*

Empirical research specifically on the macroeconomic determinants of VN-Index returns remains limited relative to the broader emerging market literature. Two studies are most directly relevant to the present paper.

[9] examine monetary policy transmission in Vietnam using a VAR model on monthly data from January 2003 to December 2012, focusing on the interest rate, exchange rate, and asset price chan-

nels. Their key finding for the asset price channel is that while CPI Granger-causes VN-Index – reflecting the impact of inflation surprises on equity valuations – VN-Index does not significantly affect CPI, implying a one-directional transmission relationship. Regarding the interest rate channel, [9] document evidence of a *cost channel*: increases in the interbank rate (VNIBOR) raise firm borrowing costs and push up prices rather than reducing output and inflation through the conventional demand channel. They attribute the weakness of the exchange rate channel to the State Bank of Vietnam's tight administrative control over the USD/VND rate, which limits the pass-through of exchange rate movements to the broader economy. These findings provide the primary benchmark against which the present study's results are compared, with the key distinction that the present paper adopts the deposit rate – the most directly observable monetary signal for retail equity investors – rather than the interbank rate as the domestic monetary policy proxy.

[5] extend the analysis of VN-Index dynamics to encompass both domestic macroeconomic conditions and regional market linkages, employing a time-varying structural VAR (TVSVAR) framework on data from July 2000 to December 2016. Their key findings suggest that the easing of monetary and credit conditions, stable and stronger currency, and economic growth have played a significant and positive role in the development of the stock market in Vietnam. On the regional integration dimension, their study analyses the role of regional markets including Thailand, Japan, Hong Kong, and China in explaining the dynamics of the Vietnamese stock market, finding that regional comovements are a significant driver of VN-Index fluctuations. The time-varying framework of [5] also reveals that the VN-Index became progressively less sensitive to external shocks over the sample period, suggesting increasing market maturity and improved price discovery.

E. *Research Gaps*

Despite the growing literature on macroeconomic determinants of Vietnamese equity returns, several important gaps remain.

First, much of the existing empirical evidence does not cover recent periods characterised by unusually large global and domestic shocks, including the COVID-19 crisis and the rapid tightening of U.S. monetary policy during 2022–2023. As a result, the post-2020 behaviour of Vietnamese equity markets under heightened global financial volatility remains underexplored.

Second, the role of global monetary policy conditions, particularly U.S. interest rate dynamics, has not been fully incorporated into empirical models of Vietnamese equity returns alongside domestic macroeconomic variables. This limits the understanding of how external financial cycles interact with domestic market dynamics in an open emerging economy context.

Third, empirical approaches in this literature are often fragmented, focusing separately on causality analysis, dynamic responses, or forecasting performance, rather than integrating these elements within a unified VAR-based framework. This restricts a comprehensive evaluation of both transmission mechanisms and predictive content of macroeconomic variables.

III. DATA AND METHODOLOGY

A. *Data*

This study employs monthly data spanning January 2010 to December 2024, yielding 180 observations. The sample is chosen to encompass two major structural episodes: the COVID-19 market shock of March 2020, which produced the largest single-month decline in VN-Index history, and the Federal Reserve's 2022–2023 rate-hiking cycle, the most aggressive U.S. monetary tightening in four decades.

Variable selection is guided by the Arbitrage

Pricing Theory of [2], under which macroeconomic variables represent systematic risk factors that generate undiversifiable risk and are therefore priced into equity returns. Six variables are included, classified by their economic role and described in Table 1. The VN-Index (VNI) serves as the dependent variable, representing the benchmark equity return of the Ho Chi Minh Stock Exchange. Consumer price inflation (CPI) is included because unexpected inflation erodes the real value of corporate cash flows and raises uncertainty over future earnings [2, 4]. The deposit rate (DEP) is included as a proxy for domestic monetary conditions and the opportunity cost of equity investment. An increase in deposit rates raises the relative attractiveness of risk-free savings instruments compared with equities, thereby reducing investor demand for stocks and placing downward pressure on VN-Index valuations. This channel is particularly relevant in the Vietnamese equity market, where retail investors account for the majority of trading activity and are highly sensitive to changes in deposit returns. The nominal effective exchange rate (NEER) captures the competitiveness channel: exchange rate movements alter the relative profitability of internationally exposed firms and influence investor sentiment toward the domestic currency [5, 7]. The global oil price (OIL) represents supply-side cost shocks that compress corporate profit margins and reduce expected dividends [6]. Finally, the U.S. Federal Funds Rate (FFR) is included as the key external monetary policy variable. [7] demonstrate that international economic forces are priced into equity markets even under market segmentation, while [3] show that unexpected FFR changes shift the global equity risk premium through capital flow and risk appetite channels– effects that were particularly pronounced during the 2022–2023 tightening cycle [8].

Table 1: Variable Descriptions and Sources

Variable	Abbrev.	Symbol	Transform
VN-Index	VNI	VNI_t	$\ln(\text{VNI}_t)$
CPI YoY	CPI	CPI_t	Level (%)
Deposit Rate	DEP	DEP_t	Level (%)
Nominal Eff. Exchange Rate	NEER	NEER_t	$\ln(\text{NEER}_t)$
Fed Funds Rate	FFR	FFR_t	Level (%)
Oil Price (Brent)	OIL	OIL_t	$\ln(\text{OIL}_t)$

The natural logarithm is applied to VNI, NEER, and OIL to stabilise variance and ensure that subsequent first differences yield interpretable percentage changes; CPI, DEP, and FFR are retained in level form as they are already expressed in percentage terms. Whether first differencing is required for these variables is determined empirically in Section B.

Descriptive statistics for the transformed variables in levels are reported in Table 2. CPI exhibits pronounced right-skewness (skewness = 2.18), driven by the inflation spike of 2011 (peak: 23.0%), followed by a sustained disinflation trend to below 5% from 2013 onward. FFR displays similarly high skewness (1.48), reflecting the decade-long near-zero rate environment of 2010–2021 and the abrupt hiking cycle of 2022–2023 during which the Federal Funds rate rose from 0.05% to 5.33%. These distributional features motivate the use of log transformation and first differencing, the necessity of which is formally established in the methodology section.

B. Methodology

The empirical procedure follows the standard VAR framework of [12], proceeding through stationarity testing, cointegration analysis, lag selection, model estimation and adequacy checking, dynamic analysis, and out-of-sample forecasting. The VAR treats all variables as jointly endogenous

Table 2: Descriptive Statistics – Transformed Variables (2010M01–2024M12)

Stat	ln(VNI)	CPI (%)	DEP (%)	ln(NEER)	FFR (%)	ln(OIL)
N	180	180	180	180	180	180
Mean	6.62	5.07	8.51	10.00	1.23	4.23
Std	0.41	4.66	2.32	0.07	1.72	0.33
Min	5.86	−0.98	5.90	9.80	0.05	2.81
p25	6.23	2.55	7.20	9.95	0.09	3.99
Median	6.62	3.50	7.30	10.02	0.19	4.29
p75	6.97	5.95	9.00	10.05	1.82	4.49
Max	7.31	23.02	14.00	10.10	5.33	4.74

and imposes no a priori restrictions on the structural relationships between variables. Because the VAR expresses each dependent variable solely in terms of predetermined lagged values, it constitutes a reduced-form model that is well suited to characterising the dynamic interactions among macroeconomic variables and equity returns.

1) Stationarity Testing

The VAR framework requires all variables entering the system to be stationary $I(0)$. A clearly important argument that arises when dealing with stock market data is whether to use levels or first differences in the VAR. The majority view, as highlighted by [13] and [14], is that stationary data should be used, since non-stationary data can lead to spurious regression results. Accordingly, stationarity is assessed using three complementary procedures with opposing null hypotheses to reduce the risk of reliance on any single test.

Augmented Dickey-Fuller (ADF) test. The ADF regression with constant is:

$$\begin{gathered}\Delta y_t = \alpha + \gamma y_{t-1} + \sum_{i=1}^{k} \delta_i \Delta y_{t-i} + \varepsilon_t, \\ H_0 : \gamma = 0 \quad \text{(unit root)}\end{gathered} \tag{1}$$

The lag order k is selected by AIC with a maximum of 12. Rejection of H_0 at the 5% level provides evidence of stationarity.

Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test. The KPSS procedure reverses the null: H_0 states that the series *is* stationary, based on a Lagrange Multiplier score testing principle [15]. A variable is confirmed $I(0)$ only when ADF rejects H_0 and KPSS fails to reject H_0, thereby reducing the risk of spurious conclusions from either test alone.

Phillips–Perron (PP) test. Where ADF and KPSS yield conflicting or inconclusive results, the PP test is applied. This test uses a non-parametric correction for serial correlation and heteroskedasticity in the regression residuals [16], making it more robust than ADF in the presence of structural breaks or time-varying variance.

Results. Applied to the level series, all six variables fail the joint ADF/KPSS stationarity condition – ADF fails to reject the unit root null and KPSS rejects the stationarity null – confirming that all variables are $I(1)$. The Johansen cointegration test is therefore applied before first differencing to determine the appropriate estimation framework, as described in the following subsection.

2) Johansen Cointegration Test

Since all six variables are confirmed $I(1)$ in levels, the Johansen procedure is applied to test whether a long-run cointegrating relationship exists among them. If cointegration is detected, a

Vector Error Correction Model (VECM) is required to preserve long-run information; absent cointegration, a VAR on first differences is appropriate. It has been argued that, even in the presence of cointegration, unrestricted VARs perform well at short horizons [17, 18], and [19] demonstrates that the impulse response results from unrestricted VARs and VECMs are nearly identical in short-run analysis. Nevertheless, the cointegration test is carried out as a formal specification check.

The Johansen procedure tests the rank r of the long-run multiplier matrix sequentially using two complementary statistics. Both the Trace statistic ($84.22 < 95.75$ at the 5% level) and the Maximum-Eigenvalue statistic ($33.39 < 40.08$ at the 5% level) fail to reject the null of zero cointegrating vectors at the first step of the sequential procedure, indicating the absence of a long-run equilibrium relationship among the six variables. A VAR on first differences is therefore appropriate, and a VECM is not required.

Given these results, first differences are applied to all variables. To confirm that first differencing achieves stationarity, ADF and KPSS tests are re-applied to all differenced series. Five variables are straightforwardly confirmed $I(0)$, with ADF p-values ranging from 0.000 to 0.005 and KPSS statistics falling uniformly below the 5% critical value. For ΔFFR, the ADF test yields an inconclusive result ($p = 0.096$), attributable to the structural break between the near-zero rate regime of 2010–2021 and the aggressive hiking cycle of 2022–2023, which reduces the test's power to distinguish a unit root from a stationary process with a level shift. The PP test resolves this ambiguity by rejecting the unit root null ($p = 0.000$), confirming that ΔFFR is $I(0)$. All six first-differenced variables are therefore confirmed stationary and enter the system vector:

$$\boldsymbol{y}_t = \begin{pmatrix} \Delta\mathrm{FFR}_t,\ \Delta\ln\mathrm{OIL}_t,\ \Delta\mathrm{CPI}_t, \\ \Delta\mathrm{DEP}_t,\ \Delta\ln\mathrm{NEER}_t,\ \Delta\ln\mathrm{VNI}_t \end{pmatrix}' \tag{2}$$

where $\Delta x_t = x_t - x_{t-1}$ denotes the first difference and $\Delta \ln x_t \approx$ the percentage change in x_t.

3) *Optimal Lag Length Selection*

The appropriate number of lags is an important specification decision in any VAR analysis. To determine the optimal lag length p, both the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) are computed for VAR models with lag orders ranging from $p = 0, 1, \ldots, 12$:

$$\mathrm{AIC}(p) = \ln\left|\hat{\boldsymbol{\Sigma}}_u(p)\right| + \frac{2}{T}pk^2 \tag{3}$$

$$\mathrm{BIC}(p) = \ln\left|\hat{\boldsymbol{\Sigma}}_u(p)\right| + \frac{\ln T}{T}pk^2 \tag{4}$$

where $\hat{\boldsymbol{\Sigma}}_u(p)$ denotes the estimated residual covariance matrix at lag order p, $k = 6$ is the number of endogenous variables, and $T = 177$ is the effective sample size after differencing and lag adjustment.

BIC is prioritised because it applies a stronger penalty for adding extra lags, which generally leads to simpler and more stable models in relatively small samples. BIC selects $p_0 = 1$, while AIC selects $p_0 = 2$. To assess whether one lag is sufficient, the multivariate Portmanteau test is applied to the residuals of the VAR(1) model. The null hypothesis of no residual autocorrelation is rejected ($p = 0.021 < 0.05$), implying that one lag is insufficient to capture the dynamic structure of the system. Increasing the lag length to $p_1 = 2$ eliminates this problem: the Portmanteau test fails to reject the null ($p = 0.143 > 0.05$), confirming that VAR(2) residuals behave as white noise. The final specification adopted throughout this study is therefore VAR(2).

4) VAR Specification

The reduced-form VAR(2) model estimated on first-differenced variables is:

$$\boldsymbol{y}_t = \boldsymbol{c} + \mathbf{A}_1 \boldsymbol{y}_{t-1} + \mathbf{A}_2 \boldsymbol{y}_{t-2} + \boldsymbol{u}_t, \quad \text{where} \quad \boldsymbol{u}_t \sim (\mathbf{0}, \boldsymbol{\Sigma}_u) \tag{5}$$

where $\boldsymbol{c}$ is a (6×1) intercept vector; $\mathbf{A}_1$ and $\mathbf{A}_2$ are (6×6) matrices of autoregressive coefficients; and $\boldsymbol{u}_t$ is the vector of reduced-form residuals with zero mean and covariance matrix $\boldsymbol{\Sigma}_u$. Each of the six equations is estimated by OLS, which yields consistent and asymptotically efficient estimates under stability and white-noise residuals.

The VN-Index equation of primary interest is:

$$\begin{aligned}\Delta \ln \mathrm{VNI}_t &= c_1 + a_{11}^{(1)} \Delta \ln \mathrm{VNI}_{t-1} + a_{12}^{(1)} \Delta \mathrm{CPI}_{t-1} \\ &+ a_{13}^{(1)} \Delta \mathrm{DEP}_{t-1} + a_{14}^{(1)} \Delta \ln \mathrm{NEER}_{t-1} \\ &+ a_{15}^{(1)} \Delta \mathrm{FFR}_{t-1} + a_{16}^{(1)} \Delta \ln \mathrm{OIL}_{t-1} \\ &+ a_{11}^{(2)} \Delta \ln \mathrm{VNI}_{t-2} + a_{12}^{(2)} \Delta \mathrm{CPI}_{t-2} \\ &+ a_{13}^{(2)} \Delta \mathrm{DEP}_{t-2} + a_{14}^{(2)} \Delta \ln \mathrm{NEER}_{t-2} \\ &+ a_{15}^{(2)} \Delta \mathrm{FFR}_{t-2} + a_{16}^{(2)} \Delta \ln \mathrm{OIL}_{t-2} + u_{1t}\end{aligned} \tag{6}$$

The five remaining equations follow an identical structure with the same set of regressors.

5) Model Adequacy Checks

Several post-estimation diagnostic tests are conducted to verify that the VAR(2) specification satisfies the assumptions required for valid impulse response and variance decomposition analysis, with results summarised in Table 3. The Portmanteau test is evaluated at lag 12, consistent with the monthly frequency of the data and the convention of testing up to one full seasonal cycle [20]. The ARCH-LM test uses $q = 5$ lags, sufficient to detect short-run volatility clustering typical of financial time series.

Table 3: Model Adequacy Diagnostics – VAR(2)

Test	Null Hypothesis	Result	Conclusion
Stability	All roots inside unit circle	Max = 0.756	Pass
Portmanteau (lag 12)	No serial correlation in residuals	$p = 0.143$	Pass
Jarque–Bera	Residuals are normally distributed	$p = 0.000$	Fail[a]
ARCH-LM ($q = 5$)	Constant variance in residuals	DEP, OIL fail	Noted[b]
Residual correlation	Shocks are uncorrelated across equations	FFR–OIL: $r = 0.44$	Cholesky applied

[a] Residuals are non-normal due to the extreme COVID-19 market shock of March 2020.
[b] ARCH effects are detected in the DEP and OIL equations only.

The Jarque–Bera test rejects residual normality, primarily due to the extreme decline in VN-Index returns during the COVID-19 market crash of March 2020. Although OLS estimates remain consistent under non-normal residuals [**?**], the asymptotic confidence bands for impulse responses should nevertheless be interpreted with caution.

ARCH-LM tests further detect time-varying residual variance in the DEP and OIL equations, consistent with the volatility clustering commonly observed in interest rate and commodity markets [**?**]. By contrast, the VNI equation itself shows no evidence of ARCH effects ($p = 0.992$), suggesting that the main impulse response results for VN-Index remain reasonably reliable. However, residual heteroskedasticity may still cause asymptotic IRF confidence bands to understate true uncertainty during highly volatile periods [20].

Finally, the residual correlation matrix reveals several non-negligible contemporaneous correlations, most notably between FFR and OIL ($r = 0.439$), as well as between VNI and OIL ($r = 0.301$) and between VNI and FFR ($r = 0.244$). These results confirm that reduced-form shocks are correlated across equations within the same period, thereby motivating the use of Cholesky orthogonalisation for structural identification in the impulse response analysis discussed in Section 6).

6) Dynamic Analysis

Granger Causality. For each predictor x_{jt}, the pairwise Granger causality test evaluates whether past values of x_j improve the prediction of $\Delta \ln \mathrm{VNI}_t$ beyond its own history. The null hypothesis is:

$$H_0 : \ a_{1j}^{(1)} = a_{1j}^{(2)} = 0 \tag{7}$$

tested via an F-statistic at the two-lag horizon consistent with the VAR(2) specification. Rejection at the 5% level indicates that x_j Granger-causes $\Delta \ln \mathrm{VNI}_t$. As emphasised by [21], this reflects predictive precedence rather than structural causality.

Orthogonalized Impulse Response Functions (OIRF). Because the reduced-form residuals are contemporaneously correlated ($\hat{\Sigma}_u \neq I$), structural identification is achieved via Cholesky decomposition:

$$\hat{\Sigma}_u = PP', \qquad P \text{ is lower triangular} \tag{8}$$

The OIRF of VN-Index at horizon h to a unit structural shock in variable j is:

$$\mathrm{OIRF}(h, j) = \Phi_h \, p_j \tag{9}$$

where Φ_h is the h-step moving average matrix and p_j is the j-th column of P. The Cholesky ordering adopted is:

$$\Delta\mathrm{FFR}_t \ \to \Delta \ln \mathrm{OIL}_t \ \to \ \Delta\mathrm{CPI}_t \ \to \\ \Delta\mathrm{DEP}_t \ \to \ \Delta \ln \mathrm{NEER}_t \ \to \ \Delta \ln \mathrm{VNI}_t \tag{10}$$

This ordering places global variables first on the grounds that U.S. monetary policy and international commodity prices are contemporaneously exogenous to the Vietnamese economy within a given month [7]. VN-Index is ordered last because financial asset prices are assumed to adjust instantaneously to all available information, a standard assumption in the asset pricing literature [3]. Confidence bands are computed at the 90% level using asymptotic standard errors over a 24-month horizon.

Forecast Error Variance Decomposition (FEVD). The FEVD quantifies the percentage of the h-step VN-Index forecast error variance attributable to each structural shock:

$$\omega_j(h) = \frac{\sum_{s=0}^{h-1} (\Phi_s \, p_j)_1^2}{\mathrm{MSE}_h(\Delta \ln \mathrm{VNI}_t)} \times 100\%, \\ \text{where} \quad \sum_{j=1}^{6} \omega_j(h) = 100\% \tag{11}$$

FEVD is computed at horizons $h \in \{1, 3, 6, 12, 18, 24\}$ months to capture both short-run and medium-run dynamics.

7) Forecasting and Forecast Evaluation

Out-of-sample evaluation. Prior to generating forward-looking forecasts, the predictive accuracy of the VAR(2) model is assessed through a pseudo-out-of-sample evaluation. Following standard practice [22], the sample is split into a training period (2010M02–2023M12, 167 observations) and a hold-out period (2024M01–2024M12, 12 observations). The VAR(2) is estimated on the training sample and used to generate a 12-step ahead forecast sequence, which is then compared against the actual VN-Index values observed in 2024. Forecast accuracy is assessed using four metrics:

$$\mathrm{RMSE} = \sqrt{\frac{1}{H} \sum_{h=1}^{H} \left(P_{T+h} - \hat{P}_{T+h}\right)^2} \tag{12}$$

$$\mathrm{MAE} = \frac{1}{H} \sum_{h=1}^{H} \left| P_{T+h} - \hat{P}_{T+h} \right| \tag{13}$$

$$\mathrm{MAPE} = \frac{1}{H} \sum_{h=1}^{H} \left| \frac{P_{T+h} - \hat{P}_{T+h}}{P_{T+h}} \right| \times 100\% \tag{14}$$

$$U = \frac{\mathrm{RMSE}_{\mathrm{VAR}}}{\mathrm{RMSE}_{\mathrm{RW}}} \tag{15}$$

where $H = 12$, P_{T+h} denotes the actual VN-Index level, $\hat{P}_{T+h}$ the VAR(2) forecast, and $\hat{P}_{T+h}^{\mathrm{RW}} =$

P_{T+h-1} is the naïve random walk benchmark. A Theil $U < 1$ indicates that the VAR(2) outperforms the random walk. Evaluation results are reported in Section IV.

12-month ahead forecast. Having validated the model's predictive performance, the full sample (2010M02–2024M12, 179 observations) is used to re-estimate the VAR(2) and generate a 12-month ahead point forecast for the period 2025M01–2025M12. The h-step ahead point forecast is:

$$\hat{\boldsymbol{y}}_{T+h|T} = \hat{\boldsymbol{c}} + \hat{\mathbf{A}}_1 \, \hat{\boldsymbol{y}}_{T+h-1|T} + \hat{\mathbf{A}}_2 \, \hat{\boldsymbol{y}}_{T+h-2|T} \quad (16)$$

where $\hat{\boldsymbol{y}}_{T+j|T} = \boldsymbol{y}_{T+j}$ for $j \leq 0$. Forecasts of $\Delta \ln \mathrm{VNI}_t$ are converted to VN-Index levels via:

$$\hat{P}_{T+h} = P_T \cdot \exp\Bigl(\sum_{s=1}^{h} \widehat{\Delta \ln \mathrm{VNI}}_{T+s}\Bigr) \quad (17)$$

A 90% prediction interval is derived from the h-step forecast error covariance matrix. Forecast results are presented in Section IV.

IV. EMPIRICAL RESULTS

A. *Granger Causality*

Table 4 reports the pairwise Granger causality test results for the null hypothesis that each macroeconomic variable does not Granger-cause $\Delta \ln \mathrm{VNI}_t$, evaluated at the two-lag horizon consistent with the VAR(2) specification.

The results reveal that ΔFFR is the only variable that Granger-causes $\Delta \ln \mathrm{VNI}_t$ at the 5% significance level ($F = 4.395$, $p = 0.014$). The joint F-test on both lags confirms that past changes in the FFR contain incremental predictive information for VN-Index returns beyond the index's own history. In contrast, none of the domestic variables – CPI, Deposit Rate, or NEER – nor the global oil price exhibits statistically significant predictive power at conventional significance levels.

This finding is consistent with [3], who argue that U.S. monetary policy affects equity markets primarily through the equity risk premium channel. A tightening of the Federal Funds Rate increases the required return on risky assets by raising perceived financial risk and reducing investors' willingness to bear risk, thereby lowering equity valuations. For emerging markets such as Vietnam, tighter U.S. monetary policy may also weaken global risk appetite and reduce capital flows toward riskier emerging-market assets, placing downward pressure on the VN-Index.

The absence of Granger causality from domestic variables does not necessarily imply that these variables have no effect on the VN-Index. As noted by [5], emerging market equity prices tend to respond to macroeconomic shocks contemporaneously rather than with a lag, so their influence may be captured in the impulse response analysis rather than in the Granger causality test.

B. *Impulse Response Analysis*

Figure 1 presents the orthogonalized impulse response functions (OIRFs) of $\Delta \ln \mathrm{VNI}_t$ to a one-standard-deviation structural shock in each of the five macroeconomic variables over a 24-month horizon. The shaded bands represent 90% asymptotic confidence intervals.

Response to FFR shock. A one-standard-deviation increase in ΔFFR (≈ 0.116 percentage points) generates a statistically significant negative response in $\Delta \ln \mathrm{VNI}_t$ at month 1, with the confidence interval lying entirely below zero. This suggests that an unexpected tightening of U.S. monetary policy reduces Vietnamese equity returns in the short run. The effect fades quickly and becomes statistically insignificant after month 2, implying that the VN-Index absorbs monetary policy information relatively rapidly. The result is consistent with [3], who show that equity markets respond strongly to monetary policy surprises through changes in investor risk appetite and re-

Table 4: Pairwise Granger Causality Tests: $X \not\to \Delta \ln \text{VNI}$

Predictor (X)	F (lag 1)	p (lag 1)	F (lag 1&2)	p (lag 1&2)	Decision
ΔFFR	7.938	0.005	4.395	0.014	Granger-causes VNI **
$\Delta \ln$ OIL	1.922	0.167	0.895	0.411	No evidence
ΔCPI	0.577	0.449	0.346	0.708	No evidence
ΔDEP	0.789	0.376	0.426	0.654	No evidence
$\Delta \ln$ NEER	0.281	0.597	0.179	0.836	No evidence

Note: F (lag 1) is the F-statistic for the first lag only; F (lag 1&2) is the joint test on both lags, consistent with the VAR(2) specification. Significance: ** $p < 0.05$. H_0: X does not Granger-cause $\Delta \ln \text{VNI}_t$.

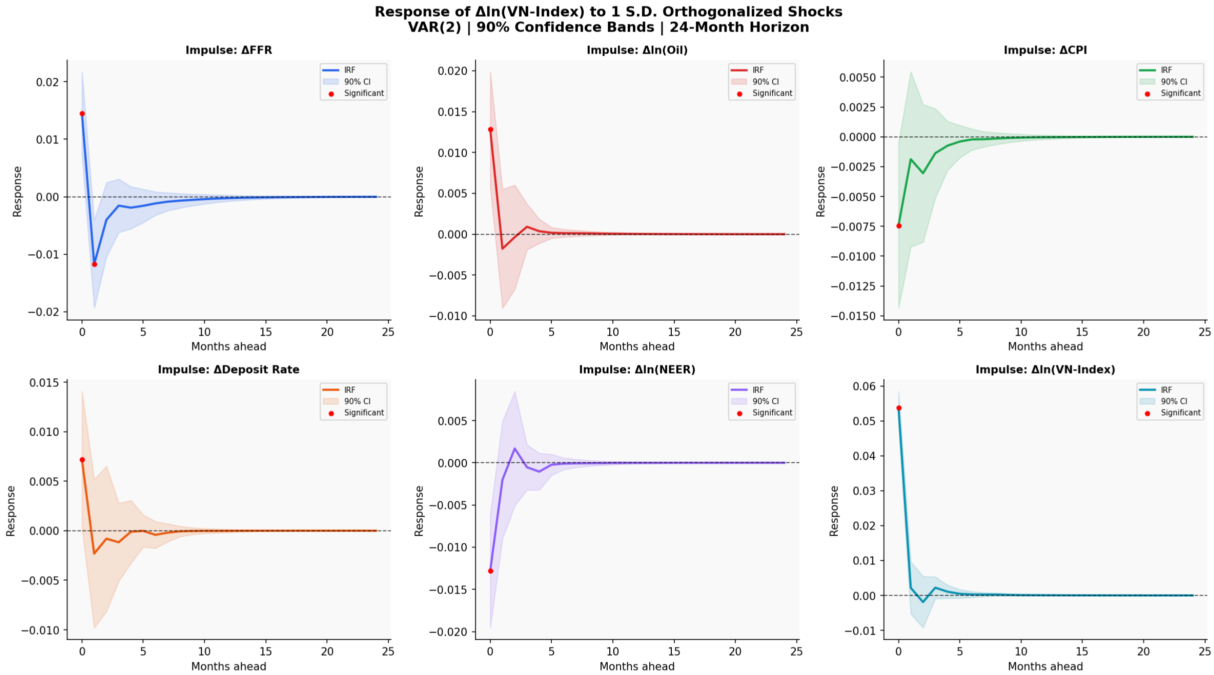

Figure 1: Orthogonalized Impulse Responses of $\Delta \ln \text{VNI}_t$ to Macroeconomic Shocks.
Note: Shaded bands = 90% asymptotic CI. Filled dots indicate months where the CI does not include zero (statistically significant response).

quired returns on risky assets.

Response to CPI shock. A positive inflation shock produces a negative response in VN-Index returns, consistent with the view that higher inflation raises discount rates and weakens real corporate profitability [2]. However, the confidence interval includes zero at all horizons, indicating that the effect is not statistically significant within the VAR(2) framework. This suggests that inflationary pressures affect the stock market mainly through indirect or longer-run channels.

Response to Deposit Rate shock. A positive shock to the deposit rate leads to a negative response in VN-Index returns, consistent with the idea that higher deposit rates increase the attractiveness of risk-free savings relative to equities. Nevertheless, the response remains statistically insignificant, implying that the short-run transmission of domestic interest rate shocks to equity returns is limited during the sample period.

Response to NEER shock. A depreciation shock in the nominal effective exchange rate generates a negative response in VN-Index returns, consistent with concerns over higher import costs and capital outflow pressures. However, the response is not statistically significant at any horizon, possibly reflecting the relatively managed nature of Vietnam's exchange rate regime [9].

Response to Oil Price shock. The response of VN-Index returns to an oil price shock is mildly negative, reflecting the adverse cost effects of rising oil prices on an oil-importing economy. However, the effect is statistically insignificant throughout the horizon, suggesting that global oil price fluctuations do not exert a strong short-run influence on Vietnamese equity returns once other macroeconomic variables are taken into account.

C. *Forecast Error Variance Decomposition*

Table 5 presents the FEVD of $\Delta \ln \mathrm{VNI}_t$ at selected horizons from 1 to 23 months. The decomposition quantifies the percentage of VN-Index forecast error variance attributable to each structural shock, with each row summing to 100%. Three findings emerge from the FEVD results.

First, own shocks account for the dominant share of VN-Index variance at all horizons, ranging from 78.3% at $h = 1$ to 77.4% at $h = 12$ and beyond. This large idiosyncratic component reflects the influence of market-specific factors–investor sentiment, momentum trading, and herd behaviour–that are not captured by the macroeconomic variables in the system. This finding is broadly consistent with the emerging market literature, which documents that domestic equity markets in developing economies are significantly driven by local investor dynamics [5].

Second, among the macroeconomic variables, FFR makes the largest contribution to VN-Index variance, rising from 9.4% at $h = 1$ to 10.0% at $h = 12$. The gradual increase in FFR's contribution across horizons indicates that the impact of U.S. monetary policy on VN-Index variability accumulates over time, consistent with the delayed propagation of global monetary shocks to emerging market capital markets documented by [3].

Third, the decomposition converges rapidly and stabilises at approximately $h = 6$, indicating that the dynamic effects of macroeconomic shocks on VN-Index returns are largely absorbed within six months. Beyond this horizon, marginal shocks contribute negligibly to forecast error variance, consistent with the stable VAR dynamics confirmed by the eigenvalue stability check.

Domestic variables–Deposit Rate (1.6%), CPI (1.9%), and NEER (4.6%) – collectively account for only 8.1% of VN-Index variance at the 12-month horizon. While NEER and OIL each contribute approximately 4.5%, their individual contributions are similar in magnitude, suggesting that global commodity prices and exchange rate dynamics play comparable marginal roles in driving VN-Index fluctuations. The deposit rate's negligible contribution (1.6%) corroborates the Granger causality finding that domestic monetary policy transmission to equity prices through the interest rate channel remains weak over the sample period.

D. *Forecasting Performance*

1) *Out-of-Sample Evaluation*

Table 6 reports the forecast accuracy metrics for the pseudo-out-of-sample evaluation over 2024M01–2024M12, using the VAR(2) estimated on the training sample 2010M02–2023M12.

Table 6: Out-of-Sample Forecast Accuracy – VAR(2), Evaluation Period: 2024M01–2024M12

Metric	$\Delta \ln$ scale	Level scale
RMSE	0.0332	87.05 points
MAE	0.0253	82.05 points
MAPE	–	6.51%
Theil U	–	0.689

Note: Theil $U = \mathrm{RMSE}_{\mathrm{VAR}}/\mathrm{RMSE}_{\mathrm{RW}}$, where the random walk benchmark uses $\hat{P}_t^{RW} = P_{t-1}$. $U < 1$ indicates VAR outperforms the random walk.

Table 5: Forecast Error Variance Decomposition of $\Delta \ln \text{VNI}_t$ (%)

Shock Variable	$h = 1$	$h = 3$	$h = 6$	$h = 18$	$h = 23$
$\Delta \ln$ VNI	78.34	77.64	77.45	77.41	77.41
ΔFFR	9.42	9.80	9.97	10.01	10.01
$\Delta \ln$ NEER	4.55	4.58	4.60	4.59	4.59
$\Delta \ln$ OIL	4.55	4.52	4.51	4.51	4.51
ΔCPI	1.59	1.87	1.89	1.89	1.89
ΔDEP	1.55	1.59	1.59	1.59	1.59

Note: Cholesky ordering: FFR → OIL → CPI → DEP → NEER → VNI. Each row reports the percentage contribution of each structural shock to the forecast error variance of VN-Index returns at different horizons. All rows sum to 100%.

The VAR(2) model achieves a MAPE of 6.51% over the 12-month evaluation window, indicating that the average forecast error represents approximately 6.5% of the actual VN-Index level. The Theil U statistic of 0.689 is below unity, confirming that the VAR(2) outperforms the naive random walk benchmark. This result validates the model's predictive capacity and provides empirical support for the use of macroeconomic variables as predictors of VN-Index movements, at least at the medium-run horizon considered here.

The relatively modest accuracy at the level scale (RMSE of 87 points against an average VN-Index of approximately 1,230 points) is consistent with the FEVD finding that 77% of VN-Index variance is attributable to own shocks, implying limited macroeconomic predictability of short-run equity returns. This is a well-documented feature of emerging market stock return forecasting [5].

2) 12-Month Ahead Forecast (2025)

Figure 2 presents the 12-month ahead point forecast for the VN-Index over 2025M01–2025M12, generated using the VAR(2) estimated on the full sample 2010M02–2024M12. The associated 90% prediction intervals are constructed from the h-step forecast error covariance matrix and converted to index levels via cumulative exponentiation.

Table 7: VN-Index Point Forecast and 90% Prediction Interval, 2025M01–2025M12

Month	Forecast (pts)	Lower 90% PI	Upper 90% PI
2025-01	1,286.68	1,166.81	1,418.87
2025-03	1,311.77	973.58	1,767.44
2025-06	1,344.04	737.60	2,449.08
2025-09	1,371.38	556.35	3,380.42
2025-12	1,396.30	418.72	4,656.18

Note: Selected months shown for brevity. Base: December 2024 VN-Index = 1,266.78 points. The widening prediction interval reflects the accumulation of forecast uncertainty over the 12-month horizon.

The point forecast projects a gradual appreciation of the VN-Index from 1,266.78 points in December 2024 to 1,396.30 points by December 2025, representing a cumulative return of approximately +10.2%. This trajectory reflects the model's expectation of a normalising macroeconomic environment, driven primarily by the anticipated moderation in FFR following the Federal Reserve's rate-cutting cycle initiated in late 2024.

The prediction intervals widen considerably over the forecast horizon–from a range of 252 points at $h = 1$ to 4,237 points at $h = 12$–reflecting the accumulation of uncertainty in multi-step VAR forecasts. This widening is an expected feature of the linear VAR framework and is consistent with the large idiosyncratic variance component documented in the FEVD analysis. The point forecast

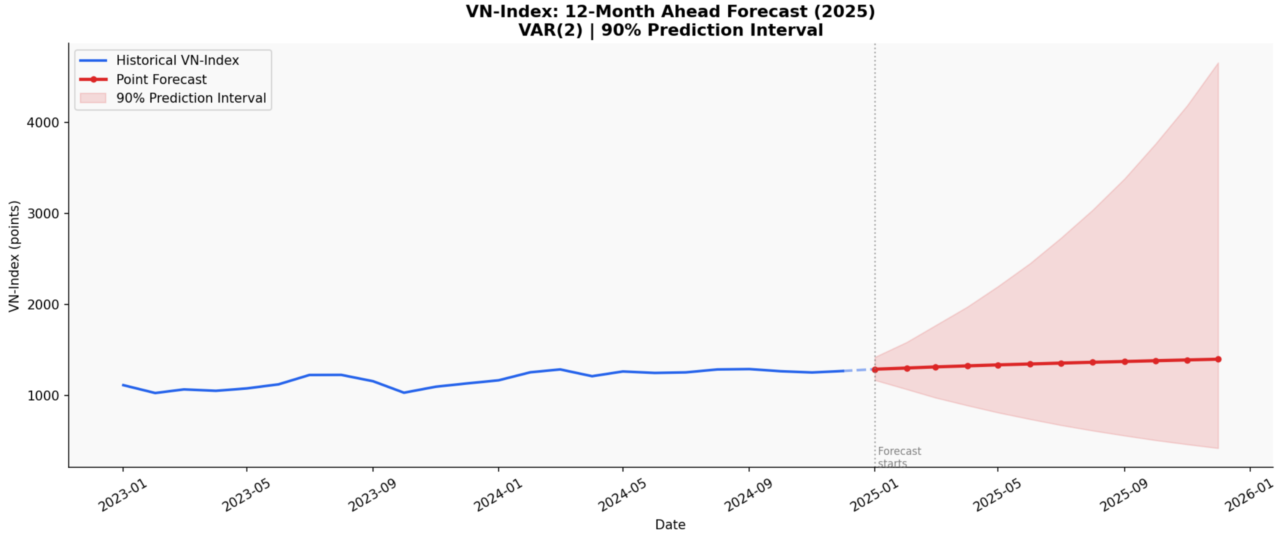

Figure 2: VN-Index 12-Month Ahead Forecast, 2025M01–2025M12. *Note:* Dashed line = point forecast. Shaded band = 90% prediction interval. Base level (December 2024): 1,266.78 points.

should therefore be interpreted as a central tendency rather than a precise prediction, and the wide prediction interval underscores the inherent difficulty of forecasting emerging market equity returns at longer horizons.

V. DISCUSSION OF FINDINGS

The empirical results yield three main findings. First, the U.S. Federal Funds Rate (FFR) is the only macroeconomic variable that Granger-causes VN-Index returns and is the dominant external driver in the system. Second, domestic monetary policy, proxied by the deposit rate, has a weak and statistically insignificant effect on equity returns. Third, VN-Index dynamics are largely driven by own shocks, which account for approximately 77% of forecast error variance at all horizons. This section interprets these results in the context of existing literature and the structural features of the Vietnamese equity market.

A. *FFR as the Main External Driver*

The finding that changes in the FFR significantly Granger-cause VN-Index returns and account for around 10% of forecast error variance is consistent with the international monetary spillover literature. U.S. monetary policy affects global equity markets primarily through the risk premium and capital flow channels [3]. A tightening of the FFR increases global discount rates and reduces the relative attractiveness of emerging market assets, leading to capital outflows and equity price declines.

For Vietnam, this transmission is mainly realised through portfolio flows rather than direct financial linkages. During major tightening cycles, foreign investors tend to reduce exposure to Vietnamese equities, contributing to downward pressure on the VN-Index. The impulse response pattern – significant in the first month and quickly dissipating – suggests that the market incorporates FFR information rapidly.

The gradual increase in the FEVD contribution of FFR over time further indicates that its effects accumulate through delayed portfolio adjustments by international investors. Overall, these results highlight the dominant role of external monetary conditions in shaping Vietnamese equity dynamics.

B. Weak Domestic Monetary Policy Channel

In contrast, the deposit rate does not Granger-cause VN-Index returns and contributes only a small fraction of forecast error variance. This suggests that the domestic interest rate channel has limited influence on equity pricing over the sample period.

One explanation is the low variability and infrequent adjustment of deposit rates, which reduces their informational content in a monthly VAR framework. In addition, Vietnam's equity market is dominated by retail investors, whose trading decisions are more strongly influenced by sentiment and momentum than by traditional discount-rate considerations.

These findings are broadly consistent with earlier studies on Vietnam's monetary transmission, which also document weak interest rate effects on financial asset prices [9]. However, the results here suggest that the opportunity cost channel of monetary policy remains underdeveloped in the equity market.

C. Dominance of Own Shocks

The most striking result is that approximately 77% of VN-Index forecast error variance is explained by its own innovations. This indicates that idiosyncratic and market-specific factors dominate macroeconomic fundamentals in driving equity returns.

This pattern is typical of emerging markets, where lower institutional participation and higher retail dominance often lead to stronger roles for sentiment and momentum. In Vietnam, trading activity is heavily retail-driven, which can amplify short-term price dynamics that are weakly linked to macroeconomic variables.

The rapid stabilisation of the impulse responses within a few months further suggests that macro shocks are short-lived, after which internal market dynamics dominate.

D. Other Macroeconomic Factors

NEER and oil prices each play relatively minor roles in explaining VN-Index variance. The weak effect of NEER is consistent with Vietnam's managed exchange rate regime, which limits exchange rate volatility and reduces its impact on equity valuations. Oil price effects are also limited, likely reflecting indirect transmission through inflation and cost channels rather than direct equity pricing mechanisms.

E. Forecasting Performance and Implications

The VAR(2) model shows reasonable out-of-sample forecasting performance, outperforming a naive benchmark. However, forecast accuracy is constrained by the large share of unexplained variance driven by own shocks.

The results suggest that macroeconomic variables – particularly the FFR – provide useful medium-term information for VN-Index forecasting, but cannot fully capture short-term market fluctuations.

F. Policy Implications

For investors, the findings imply that monitoring U.S. monetary policy is more important than focusing on domestic interest rates when assessing Vietnamese equity market conditions. FFR tightening cycles represent systematic headwinds through capital flow and risk premium channels.

For policymakers, the weak role of domestic monetary policy in equity pricing suggests limited direct influence of interest rate adjustments on stock market dynamics. External shocks transmitted through global financial conditions play a more important role. Strengthening institutional participation in the market may improve the role of macro fundamentals in price formation over time.

Overall, the results highlight the dual nature of Vietnam's equity market: strong sensitivity to global monetary conditions, combined with a dominant internal, idiosyncratic component.

VI. CONCLUSION

This study examines the macroeconomic determinants of VN-Index returns over the period 2010–2024 using a six-variable VAR(2) framework with Cholesky identification. The analysis covers major episodes of external and domestic macroeconomic stress, including the COVID-19 period and the 2022–2023 U.S. monetary tightening cycle.

Three main findings emerge. First, among all macroeconomic variables considered, only the U.S. Federal Funds Rate (FFR) is found to Granger-cause VN-Index returns. Orthogonalized impulse response analysis shows that a tightening shock in U.S. monetary policy has a statistically significant negative short-run effect on Vietnamese equities, although the impact dissipates quickly. This suggests that external monetary conditions are the primary identifiable macro-financial transmission channel affecting the Vietnamese equity market, mainly through global risk appetite and capital flow adjustments.

Second, forecast error variance decomposition indicates that VN-Index dynamics are dominated by idiosyncratic shocks. Own innovations account for approximately 77.4% of the variance at the 12-month horizon, while the FFR is the largest macroeconomic contributor at around 10.0%. Domestic macroeconomic variables, including inflation, exchange rate, oil prices, and deposit rates, jointly explain a relatively small share of fluctuations. This highlights the strong role of market-specific and sentiment-driven dynamics, consistent with the characteristics of an emerging market with high retail investor participation and limited institutional depth.

Third, out-of-sample evaluation over 2024 shows that the VAR(2) model outperforms a naive random walk benchmark, with a MAPE of 6.51% and a Theil U of 0.689. Although forecast uncertainty remains high, the results suggest that macroeconomic information—particularly U.S. monetary policy—contains useful predictive content for medium-term VN-Index movements.

Overall, the findings indicate that Vietnamese equity returns are more responsive to external monetary shocks than to domestic macroeconomic fundamentals in a systematic sense. For investors, this implies that monitoring U.S. monetary policy cycles is crucial for portfolio allocation decisions. For policymakers, the results highlight the importance of managing external spillovers, particularly through exchange rate stability and capital flow management.

Several limitations should be noted. The linear VAR framework assumes constant transmission mechanisms over time and may not fully capture structural changes associated with major macroeconomic events. Future research could extend this work using time-varying parameter VAR models or incorporate additional variables such as capital flows and investor sentiment to better capture the drivers of VN-Index dynamics.

REFERENCES

[1] Stephen A Ross. The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13(3):341–360, 1976.

[2] Nai-Fu Chen, Richard Roll, and Stephen A Ross. Economic forces and the stock market. *Journal of Business*, 59(3):383–403, 1986.

[3] Ben S Bernanke and Kenneth N Kuttner. What explains the stock market's reaction to federal reserve policy? *Journal of Finance*, 60(3):1221–1257, 2005.

[4] Eugene F Fama. Stock returns, real activity, inflation, and money. *American Economic Review*, 71(4):545–565, 1981.

[5] Muhammad Ali Nasir, Muhammad Shahbaz, Thu Thuy Mai, and Mohammad Shu-

bita. Development of vietnamese stock market: Influence of domestic macroeconomic environment and regional markets. *International Journal of Finance & Economics*, 26(1):1435–1458, 2021.

[6] Perry Sadorsky. Oil price shocks and stock market activity. *Energy Economics*, 21(5):449–469, 1999.

[7] James N Bodurtha, D Chinhyung Cho, and Lemma W Senbet. Economic forces and the stock market: An international perspective. *Global Finance Journal*, 1(1):21–46, 1989.

[8] Povilas Lastauskas and Anh DM Nguyen. Spillover effects of u.s. monetary policy on emerging markets amidst uncertainty. *Journal of International Money and Finance*, 142:103034, 2024.

[9] Xuan Vinh Võ and Phuc Canh Nguyễn. Monetary policy transmission in vietnam: Evidence from a var approach. *Journal of Asian Economics*, 34:17–27, 2014.

[10] William F Sharpe. Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, 19(3):425–442, 1964.

[11] Eugene F Fama. Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383–417, 1970.

[12] Christopher A Sims. Macroeconomics and reality. *Econometrica*, 48(1):1–48, 1980.

[13] Clive WJ Granger and Paul Newbold. Spurious regressions in econometrics. *Journal of Econometrics*, 2(2):111–120, 1974.

[14] Peter CB Phillips. Understanding spurious regressions in econometrics. *Journal of Econometrics*, 33(3):311–340, 1986.

[15] Denis Kwiatkowski, Peter CB nations Phillips, Peter Schmidt, and Yongcheol Shin. Testing the null hypothesis of stationarity against the alternative of a unit root. *Journal of Econometrics*, 54(1-3):159–178, 1992.

[16] Peter CB Phillips and Pierre Perron. Testing for a unit root in time series regression. *Biometrika*, 75(2):335–346, 1988.

[17] Robert F Engle and Byoung Soo Yoo. Forecasting and testing in co-integrated systems. *Journal of Econometrics*, 35(1):143–159, 1987.

[18] Atsuyuki Naka and David Tufte. Examining impulse response functions in cointegrated systems. *Applied Economics*, 29(12):1593–1603, 1997.

[19] Akram Maghyereh. Oil price shocks and emerging stock markets: A generalized var approach. *International Journal of Applied Econometrics and Quantitative Studies*, 1(2):27–40, 2004.

[20] Helmut Lütkepohl. *New Introduction to Multiple Time Series Analysis*. Springer, Berlin, 2005.

[21] Clive WJ Granger. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica*, 37(3):424–438, 1969.

[22] Francis X Diebold. Comparing predictive accuracy, twenty years later: A personal perspective on the use and abuse of diebold–mariano tests. *Journal of Business & Economic Statistics*, 33(1):1–9, 2015.